# Network Topology Design, Configuration and Connectivity Verification

TechBiz Academy | CCNA Internship | Week 1 – Task 1

**Lab environment:** Cisco Packet Tracer
**Topology:** 2 Routers, 2 Switches, 4 PCs
**Routing method:** Static IPv4 Routing
**Practical result:** Successful communication between all four PCs
**Supporting evidence:** CCNA Week 1 Task 1 — Notion Lab Documentation
Packet Tracker lab

## Objectives

The objective of this lab was to build and configure a functional network using Cisco Packet Tracer and understand how network devices communicate.

The practical Activities included:

1. Building network with 2 routers, 2 Switch, and 4 PCs.
2. Configuring IPv4 address on the PCs, router interface.
3. Establishing routes between the networks.
4. Testing communication with ping.
5. Tracing the ping path from PC A to PC D.
6. Understand the roles of the devices in the network.

## Network Topology

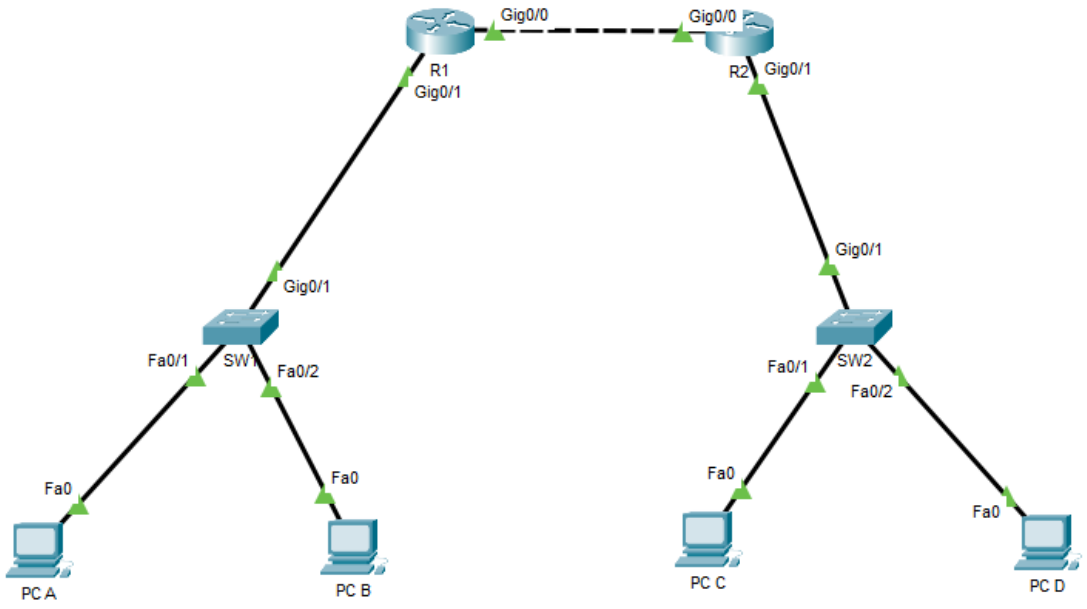



*Figure 1: Network Diagram*

The topology consists of 2 LANs 192.168.10.0 on SW1 with PC A and PC B connected to it. 192.168.20.0 network on SW2 with PC C and PC D connected to it.

The connection between R1 and R2 is 10.0.0.0/30 network.

## IP addressing

| DEVICE | INTERFACE | IP ADDRESS | SUBNET MASK |
|---|---|---|---|
| **R1** | G0/1 | 192.168.10.1 | 255.255.255.0 |
| **R1** | G0/0 | 10.0.0.01 | 255.255.255.252 |
| **R2** | G0/0 | 10.0.0.2 | 255.255.255.252 |
| **R2** | G0/1 | 192.168.20.1 | 255.255.255.0 |
| **PC A** | Fa0 | 192.168.10.10 | 255.255.255.0 |
| **PC B** | Fa0 | 192.168.10.11 | 255.255.255.0 |
| **PC C** | Fa0 | 192.168.20.10 | 255.255.255.0 |
| **PC D** | Fa0 | 192.168.20.11 | 255.255.255.0 |

## Router Configurations

R1

Router interface G0/1 was assigned IP address of 192.168.10.1 255.255.255.0 and enabled with no shutdown command. Interface G0/0 was assigned IP address of 10.0.0.1 255.255.255.252 and enabled with no shutdown.

And, static route to 192.168.20.0 network via 10.0.0.2 was configured.

```
!
!
!
!
interface GigabitEthernet0/0
 ip address 10.0.0.1 255.255.255.252
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.10.1 255.255.255.0
 duplex auto
 speed auto
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.20.0 255.255.255.0 10.0.0.2
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end

R1(config)#
```

R2

Router interface G0/1 was assigned IP address of 192.168.20.1 255.255.255.0 and enabled with no shutdown command. Interface G0/0 was assigned IP address of 10.0.0.2 255.255.255.252 and enabled with no shutdown.
And, static route to 192.168.10.0 network via 10.0.0.1 was configured.

```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#ip routin
R2(config)#ip routing
R2(config)#int g0/0
R2(config-if)#ip address 10.0.0.2 255.255.255.252
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#int g0/1
R2(config-if)#ip address 192.168.20.1 255.255.255.0
R2(config-if)#no shut

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#exit
R2(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
S    192.168.10.0/24 [1/0] via 10.0.0.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1

R2(config)#
```

The show ip route command confirmed that each router had static route to the remote LAN.

```
R1(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/1
L       192.168.10.1/32 is directly connected, GigabitEthernet0/1
S    192.168.20.0/24 [1/0] via 10.0.0.2

R1(config)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.1        YES manual up                    up
GigabitEthernet0/1     192.168.10.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1(config)#
```

```
R2(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

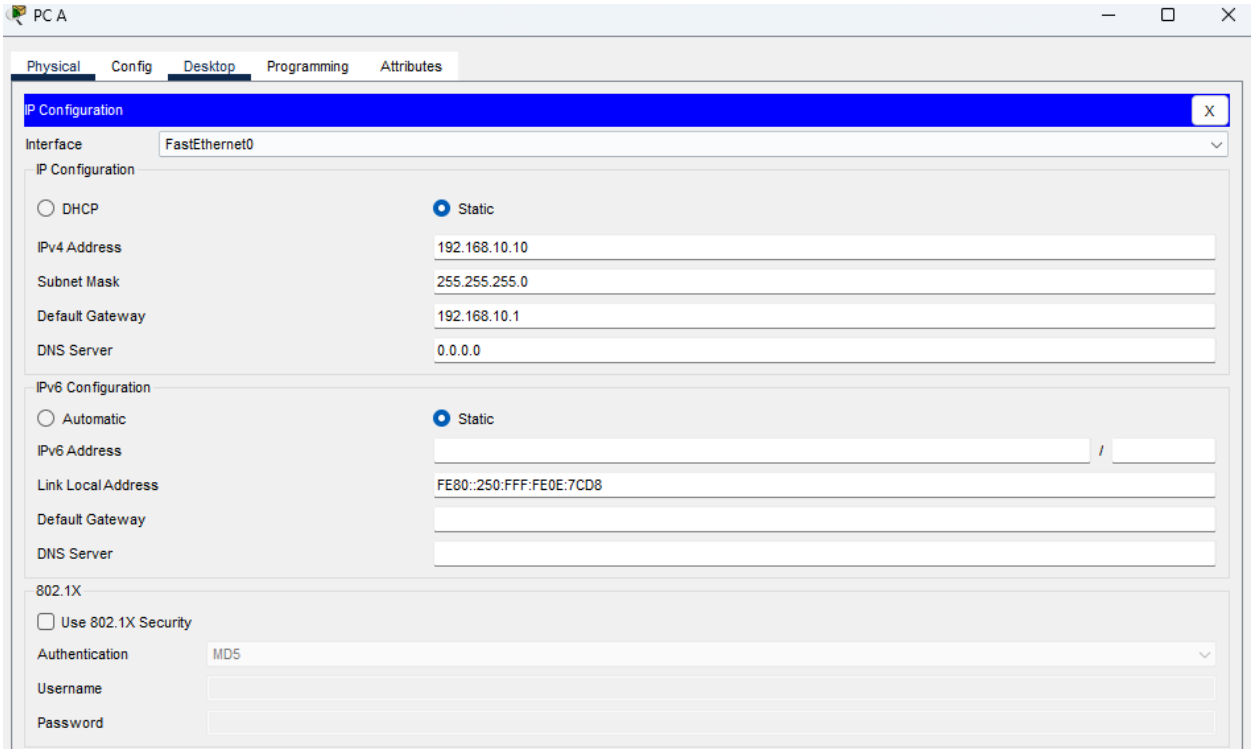
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S    192.168.10.0/24 [1/0] via 10.0.0.1
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1

R2(config)#do sh ip int br
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.0.2        YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2(config)#
```

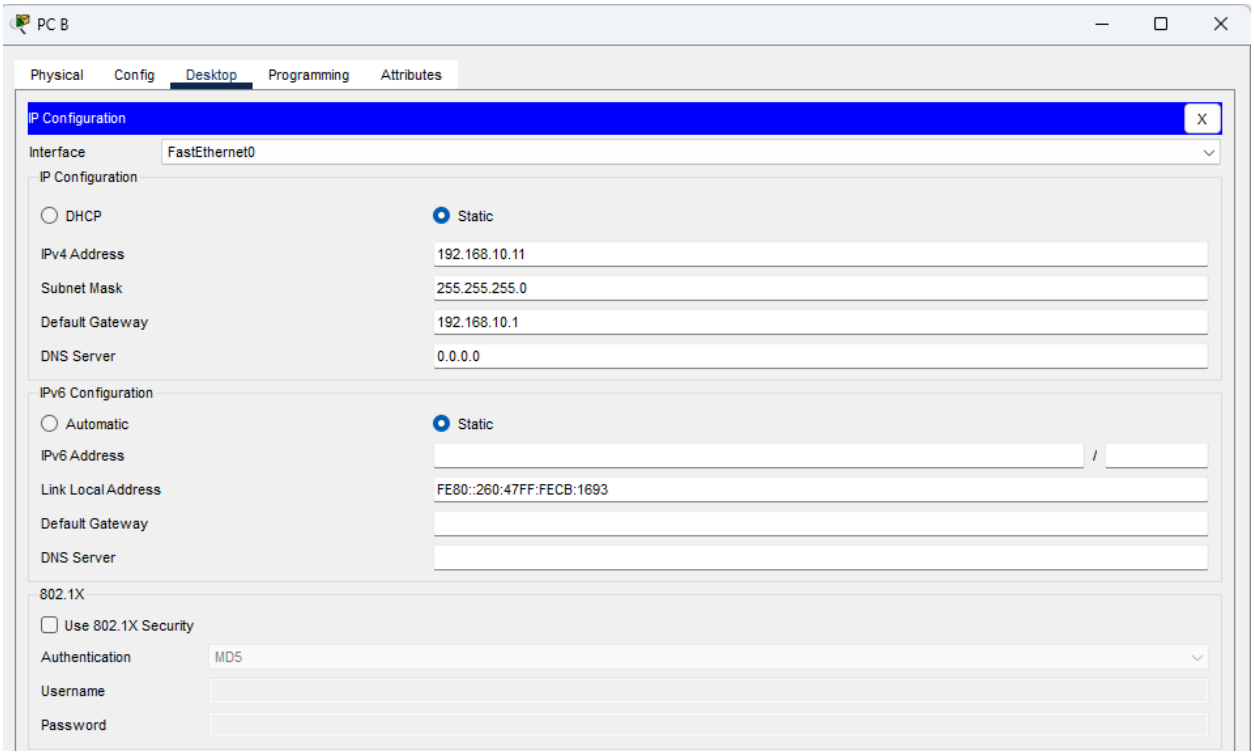
# PC configurations

Assigned the PC's with the respective IP addresses.

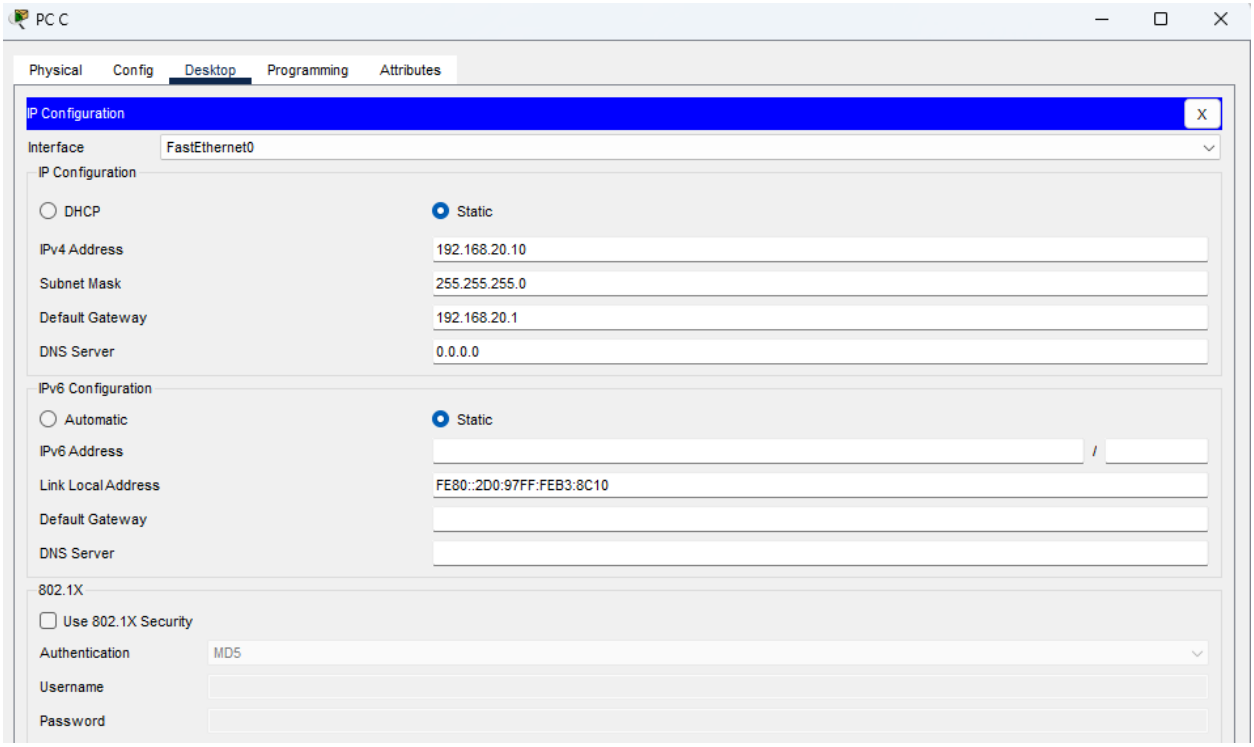
## PC A



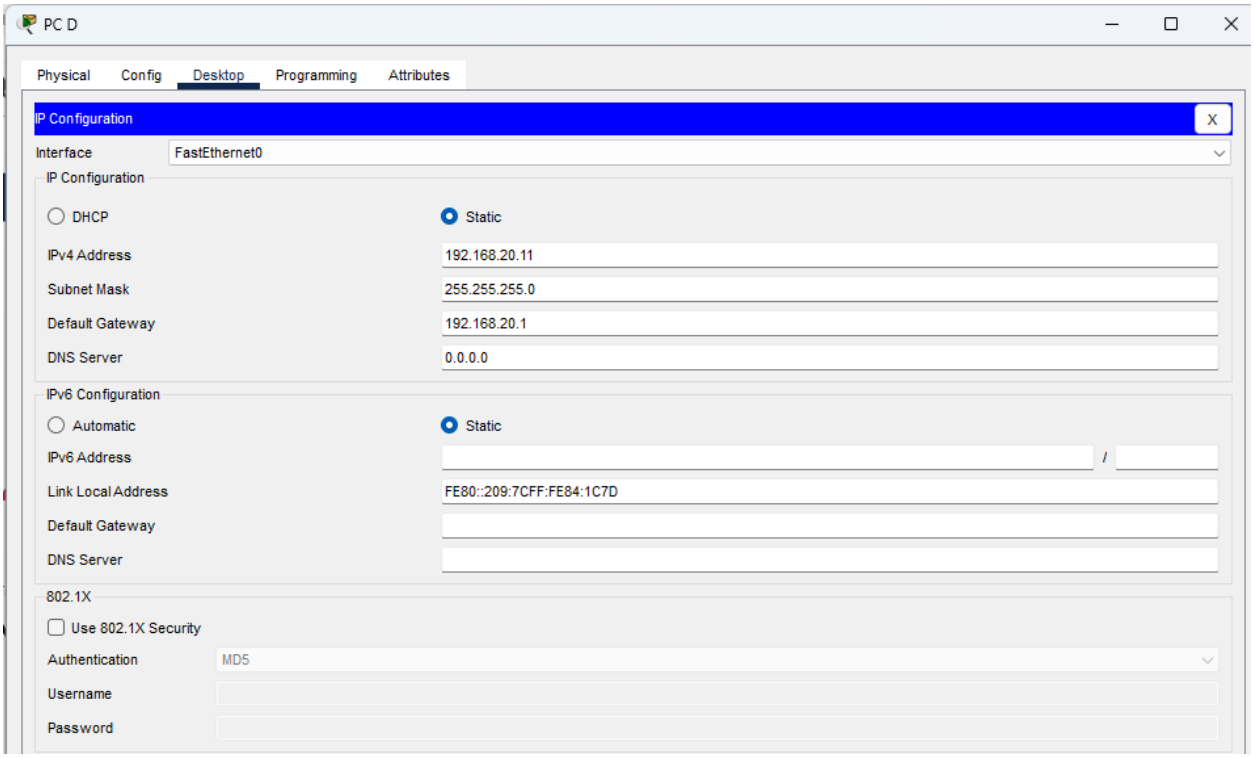

## PC B



PC C

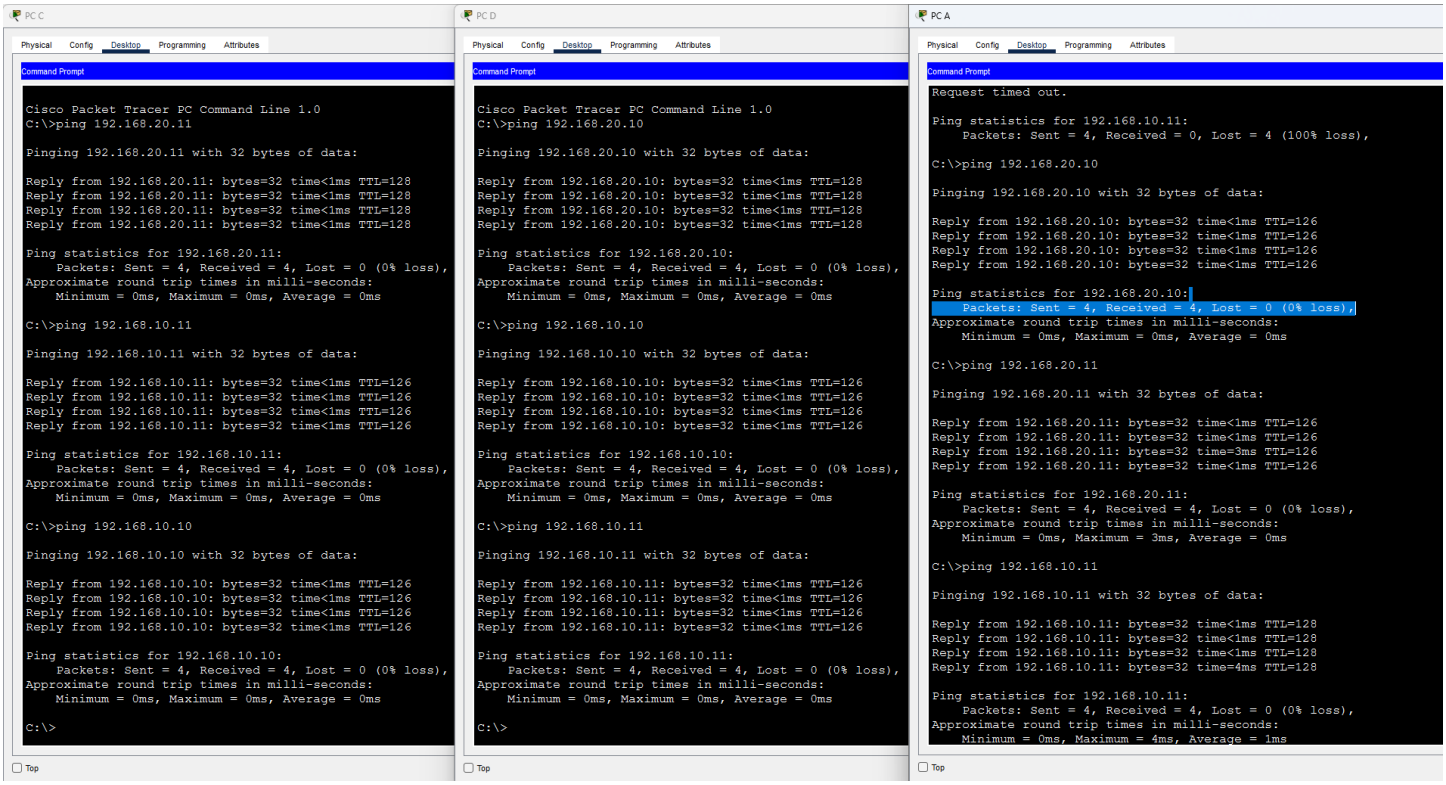

PC C
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:97FF:FEB3:8C10
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

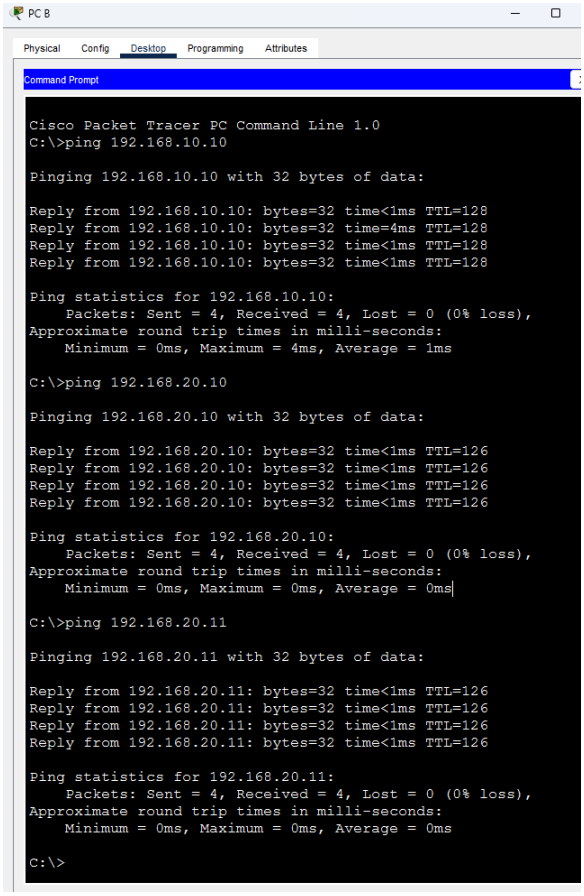
PC D


PC D
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::209:7CFF:FE84:1C7D
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

# Testing connection

Pinged all end hosts from each of the end host that resulted in success. Total: 12 successful directed PC-to-PC tests, each with four replies received and 0% packet loss in the final successful runs.



ICMP packet flow analysis

Network Device Responsibilities